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A combination of Super-resolution and Deep Learning Approaches applied to Image Forgery Detection

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Abstract— **There are a number of techniques used to create forgery images and with the case of copy-move or splicing techniques normally compatible resolutions of original and faked images are assumed in the detection. This would make the process of forgery images becoming monotony and not much challenges. In this paper, we propose an appropriate method to detect tampered images with the change in resolutions of the splicing areas that have been inserted within the images then a combination method of the superresolution approach and the deep learning technique would make an efficient method for image forgery detection. Specifically, we implement the CNN model namely VGG16 trained by the VGG network. We use the VGG16 model with 16 classes. With the test model mentioned above, the results were given with an accuracy of up to 94.64%.**

Keywords—**Image Forensics, Super-resolution, Image Forgery Detection, VGG network, Convolutional Neural Network (CNN).**

I. INTRODUCTION

In today's digital technology, the most popular method of accessing information is via the internet based on text, video, audio and, most of all, images. With such widespread phenomenon of information transmission, many people have taken advantage of tricks and tools to edit photos, falsifying information to spread to readers. The truth of the original photo is obscured by editing manipulations through Adobe Photoshop software such as Lightroom, Freehand, etc.) to distract the viewer. Looking at a big picture, this phenomenon can have a great impact on politics, economy, media, law, etc. No one can predict that the false will lead to serious consequences. This problem leads to the need to use scientific solutions to verify the authenticity of the content of the displayed image without knowing what information about the image has been faked.

One of the most common ways to counterfeit images is to embed copies of images, in other words, this is the copy-move method. The process of "embedding" an image consists of three stages: copying an image, transforming the image (in intensity or geometry, such as applying some transforms: rotate, enlarge, reduce, etc. or image resolutions may be changed), after shifting, the photo may be blurred to conceal the fake composition. Another way to falsify an editing it, and pasting it on another image. This editing is harder to detect than the first one outlined above. That's because for copy-move images, we can look up information in the image as a basis for finding the copied image area. As for the second type, it is difficult to find any information on the image as a basis for comparison. Therefore, there are many technical challenges posed to counterfeiting by image joining method.

A. Copy-Move Image Forgery Detection

SIFT [1], SURF [2] and some other reliable matching techniques [3]-[5] are used to detect copy-move forgery images. These matching methods have the ability of transition-, scale-, and rotation-. Because of that, the forged area, though different from the origin, they are still detected. However, with small subjects that contain less pattern structure detail, these techniques do not perform well as expected. Before executing a matching operation, SIFT and SURF perform an extract features process in the image base on the pattern information of objects. As a consequence, the matching stage has a tendency to overlook the duplicate area containing less pattern structure detail.

Another approach to detect copy-move forgery image is applying Discrete Cosine Transform (DCT) [6]-[8], Discrete Wavelet Transform (DWT) [9], and Zernike moment [10] to overlapping blocks that are divided from an image to extract relevant features, which are the inputs of the matching stage. In the matching stage, we compare different feature vectors as pairs. A pair of feature vector that has the similarity score lower than a predefined threshold is said to be matched. Euclidean distance is the similarity score that show how similar two vectors are.

B. Splicing forgery image detection

Splicing forgery images are more common in the real world and they are more difficult to detect than the copy-move forgery images. Zhang et al. in [11] proposed a classification model to detect such spliced objects. More clearly, moment-based feature extracted using DWT and DFT, and Image Quality Metrics were used as a feature of size 208 to train a fully connected neural network. Then, the neural network was to classify patches of a given image into two categories, namely positive and negative. However, in [12], authors introduced an application of saliency detection as a support for image splicing detection. They treated salient objects as unmodified ones, so by extracting a saliency map, they could figure our tampered regions. In a different branch, Pan et al, in [13] exploited a framework that made use of noise model to

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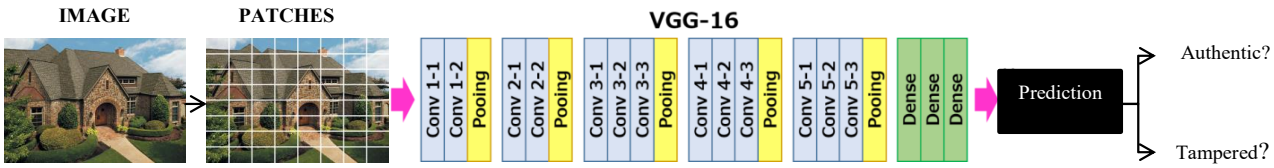

Fig.1: The overall picture of the proposed model

detect forged regions. Concretely, the input image was converted to the DCT domain. After that, a noise model was constructed to estimate noise variance in non-overlapping patches of image. Finally, a clustering stage using K-mean algorithm was designed to discriminate tampered and original patches.

C. Deep Learning-based Method

Deep Learning draws ideas from biological studies, the study of human brain processes. A large amount of data is transmitted to train an artificial neural network to learn how to best represent the pattern. Currently, the methods of using deep learning are trending and being widely applied. The reason is because of the popularity of the Internet, with massive information and we can use this large amount of data to teach neural networks to learn how to optimize how to represent a model we want. In addition, with strong support from specialized hardware such as Graphic Processing Unit (Graphic Processing Unit), training of artificial neural networks, which consumes a lot of energy for computation, is quickly deployed and to shorten the training time. Another reason is that some conventional methods using analytical mathematics, can only guarantee accuracy under certain circumstances. For deep learning, we can collect data (in the problem of identifying fake images as images) to cover all possible cases, which is still simpler than using math.

There are many ways of utilizing deep learning to assess images. Including the method of Rao and Ni [14] using a Deep Neural Network (CNN). Instead of random initialization like the following classes, weights in the first layer for CNN are initialized with 30 basic high-pass filters used in the SRM model [15] for image analysis; which is quite the highlight of this approach. The different initialization in the first layer increases accuracy, because the filters in [16] like high-pass filters, filter out unnecessary details in the image. The CNN network consists of 10 layers with an input of 128x128 image patches containing the tampering area border in the fake image. CNN has been trained and accumulated experience from test images, so it can distinguish fake and pristine images better. This method has a high accuracy of 98.04% on the standard data set CASIA v1.0 [17] 97.83% on the CASIA v2.0 and on the DVMM is 96.38%.

II. METHODOLOGY AND EXPERIMENT

The proposed method consists of two main steps, feature learning and feature extraction. Firstly, we sample labeled training images. Positive patches were sampled by sampling the corresponding masks, while negative patches were sampled by the number of active patches. In the second step, we use VGG16 model [18] to extract features based on the above patches by applying the sliding window to scan the entire image. Finally, the above-mentioned features have been synthesized for the ability to distinguish images. The reason for using VGG16 is because it has been trained on a large ImageNet dataset, which can solve our problem well.

A. VGG16 Model

VGG16 [18] is a CNN network proposed by K. Simonyan and A. Zisserman. The model after training by the VGG16 network achieved 92.7% accuracy in an ImageNet data set containing 14 million images of 1000 different classes. The VGG16 model includes multiple convolutional layers and pooling layers. VGG16's original architecture received input patches of 224x224x3 size. In our dataset, the patches were extracted from the image, 64x64x3 in size, and the output flattened to 2048 units. Convolutional layer uses 64 kernels with size 3x3, padding = 1, stride = 1, so the out of convolutional layer has the same width and height as input. After the convolutional layer, it's the pooling layer, use max poling layer with size 2x2, as a result, the width and height will decrease, but the depth will be doubled. After quite a bit of convolutional layer and pooling layer, the data is flattened into 2x2x512 and put in connected layer. From the original VGG16 model, we have removed the FNs layer and added 2 other layers (Figure 1) to serve the purpose of image differentiation, namely ReLu and sigmoid.

B. The process summary

To authenticate an image, we do a sampling with a 64x64 kernel that slides along the image to form 64x64x3 patches. These patches are then passed over to the tweaked VGG16 model so that they can be validated. Each photo is labeled 0 or 1, with 0 being authentic and 1 being Tampered. The final stage is to look at the neighborhood information to determine the respective labels of the patches. From there, it is possible to classify images in which group.

In this section, experiments were being conducted to demonstrate the effectiveness of the recommended deep learning method for detecting fake images. The model was tested on non-standard dataset, because of the change in the resolution of the tampering area of the images. However, the model's performance will be compared with the other forgery image detection methods based on the standard dataset

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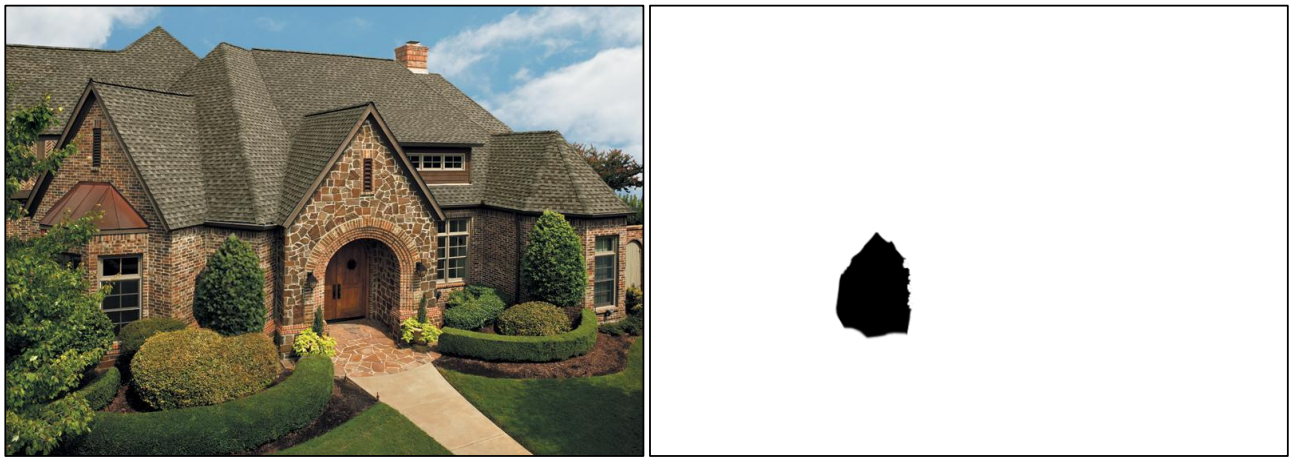

(a) (b)

Fig.2: Extracted from fake images dataset (a) Fake image, (b) The corresponding mask representing the splicing area.

C. Metrics

The experiment will use some typical metrics to assess the model. When the classification work only includes two type of images, Positive (pristine images) and Negative (tampering images), the classification output will be True Positive (TP), False Positive (FP), True Negative (TN), False Negative (FN); those are defined from the loss matrix. The metrics are computed from those 4 factors as follows.

$$Accuracy = \frac{TP+TN}{TP+TN+FN+FP} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$Fscore = \frac{2.Precison.Recall}{Precison+Recall} \quad (4)$$

TABLE I. THE LOSS MATRIX

	Predicted as Positive	Predicted as Negative
Actual: Positive	True Positive (TP)	False Negative (FN)
Actual: Negative	False Positive (FP)	True Negative (TN)

Accuracy indicates how precise the model performs. This assessment metric simply calculates the ratio between the correctly predicted samples and the total number of the samples in the test dataset, in other words, this is the overall performance of the model. However, when FN is large enough, it can lead to serious deficiencies (i.e. the model predicts fake images to be the real ones), then the Accuracy of the model will not conclude whether the model is strong enough or not. Therefore, Precision and Recall were added to overcome Accuracy's shortcomings. Specifically, Precision is defined as the ratio of Positive samples (TP) to samples predicted to be Positive (TP+FP); meanwhile, Recall shows how many Positive samples are predicted to be Positive among the real Positive ones. Depending on specific application to balance between these two metrics. Replace these two values with a single value, F-score. The higher the Accuracy, Precision, Recall, and F-score indexes, the better the model operates.

D. Image Dataset

This paper conducts the experiment on forgery images with tampering areas whose resolution has been modified. The standard dataset like CASIA [17], DVMM [23], etc.; however, only includes images with tampering areas that has a stable resolution. Hence, for model training and evaluation, a non-standard dataset [21] is created and used. We use Vandewalle's super-resolution technique [22] to change the resolution of the image before we fake it. This dataset contains 2 directories, one contains fake images and its corresponding masks and the other contains the pristine images. The mask of fake images is a grayscale image depicting the area being merged. The black pixels in the mask representing the tampering area. Figure 2 is one of the fake images and their corresponding masks taken from the used dataset.

The dataset used in this paper contains 1050 pristine images, 450 fake images and 450 corresponding masks. The images in the dataset are of random size. In fact, a color image is usually an image with 3 color channels: Red, Green, Blue (R, G, B); however, there may be times when there is a fourth color channel of yellow (Y). The images in our dataset are a combination of 1,3, and 4 color images. We divide the tested

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(a) (b)

Fig. 3: Describe the process of sampling a fake image; (a) Input Image, (b) contains 64 patches of 68 patches after sampling of a fake image in training dataset.

images in the dataset based on the number of channels. The results are shown in Table II. In the folder containing the pristine images, the majority are 3-channel color images. 4-channel color images do not have any useful information, they are simply a grid of pixels with value 0. Therefore, we will only use 1025 3-channel pristine images and remove 1, 4-channel pristine images from the dataset. Fake images are the combination of 3-channel and 4-channel images. The feature extraction used in this paper only needs information from one channel of the mask images. In the end, the dataset contains 450 fake images, so the total number of images in the dataset is 1475 images. The dataset is split into two parts: 20% for testing and 80% for training and validation. The dataset in its current state is not suitable for training the model, it must be converted into a different state that is appropriate for training. For each fake image, a corresponding mask is used to sample along the boundaries of the spliced area so as to make sure at least a 25% contribution from both spliced area and unforged area.

TABLE II.
SPLITTING THE DATASET ON CHANNEL IMAGES

channel	Pristines	Fakes	Masks
1 channel	3	0	330
3 channels	1025	386	65
4 channels	22	64	55
TOTAL	1050	450	450

These models have a distinctive boundary that only appear in fake images. The boundaries will be learned by the CNN architecture. To make the boundaries more distinct, the Otsu thresholding was used to covert grayscale mask images into binary images. Here, a kernel of 64x64 is chosen for the sampling process. Specifically, we let the kernel move (with stride = 8) along the fake image and count the number of pixels with a value of 0 in the corresponding mask. Figure 3 contains 64 patches of 68 patches after sampling of a fake image in train dataset.

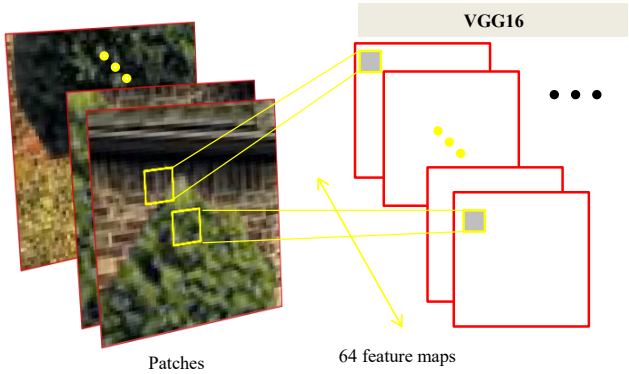

Fig. 4: The patches are input to the VGG16 model. The first layer of the network is the Convolution Layer. Performing convolution calculation is the first job of this model.

TABLE III. THE DATASET

Dataset	Authentic	Tampered
Training and Validation	175.119 patches	175.609 patches
Testing	218 images	90 images

Finally, we have 175,119 of 64x64 patches from fake images and those were labeled 1. Doing the same sample process to the pristine images, patches labeled 0 were created.

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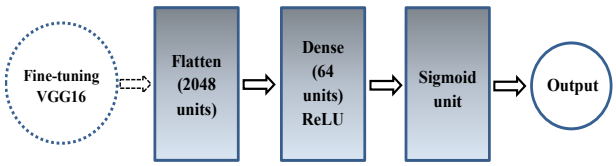
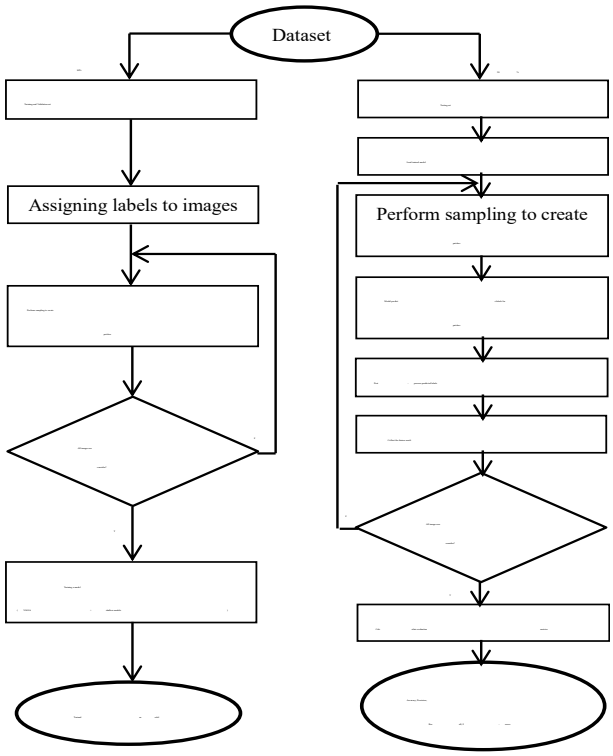


Fig. 5: The architecture of the shallow network.

In the end, there are more than 350,000 patches divided into train and validation sets, the ratio between the train set and the validation set is 6. These patches are responsible for learning the features of the network. Table III is the summary of the dataset.



(a) (b)

Fig. 6: Our flowchart model, (a) Training phase, (b) Testing phase. First, we proceed to model training. Next, in the testing process, the trained model is used to classify all patches. Finally, there is a model evaluation.

E. Training

During the training phase, the model learns how to differentiate between the Positive and Negative class using the dataset prepared above. Using the CNN network architecture in [14] and change the input patch size to 64x64x3. This network was trained on 150,000 train samples and 25,000 validation samples. The CNN network is a strong model that has more potential and improvements than some classical machine learning algorithms at the image classification task. Transfer learning appears behind this idea. Generally, a pre-trained model which was trained on a large dataset and gave out well result was used to solve this problem. In this study, the VGG16 network [18] is used for training, VGG16 is a pre-trained model that was trained with the ImageNet dataset to vectorize the images in our dataset. Firstly, we keep the ConvNet part of the VGG16 network and remove the FCs, then use the out of the remaining ConvNet as the input to the Shallow network (Fig. 5). The input is 2x2x512 tensors, with the input image being 64x64x3 also the patches (Fig. 4) we have found above. Patches are fed into the VGG16 model input to learn some features, ie patches will be passed through some convolutional and pooling layers of the VGG16 model. The model is trained by using Adam optimizer with a reduce of the custom learning rate after every 10 epochs.

$$learningrate_{new} = \frac{learningrate_{old}}{\sqrt{epoch}} \quad (5)$$
$$if\ epoch\%10\ = 0$$

The next thing to do is fine-tuning the last layers. The important thing to keep in mind here is that a pre-trained top layer model must be used for fine-tuning. The goal here is to adjust the learned weights to fit our dataset. If some randomly weights were initiated and used into the model, it would crash the learned weights of the convolution layer. A learning rate of 10^{-6} is used to fine-tune the model. Figure 6 is a summary of the modeling training that we used.

F. Testing

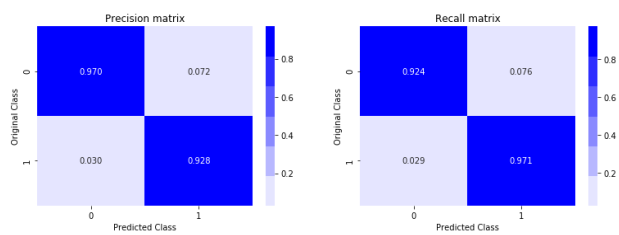


Fig. 7: This is the precision and recall matrix which was calculated from the confusion matrix.

In this section, an evaluation is carried on the test set with 218 pristine images and 90 fake images. To sample images from the test set that was created earlier, the 64x64 kernel algorithm was used with a stride of 8 (similar to the sampling process on the train set and the validation set) to get the patches of 64x64. Then, the fine-tuned VGG16 network model was used to predict the labels of patches (Fig. 6). Lastly, the model will calculate the metrics based on the loss matrix (Fig. 7) then the Table IV shows the metric of the model with the Accuracy, Precision, Recall and Fscore of the test results using pre-trained VGG16 model. It can be seen a high accuracy of the proposed method applied to image forgery detection.

TABLE IV. METRICS OF MODEL

Model	Accuracy	Precision	Recall	Fscore
VGG16-pre	*94.64%*	*92.80%*	*97.10%*	*94.91%*

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G. Comparison

In this section, we will compare the VGG16 model with other models implemented on DVMM [23] dataset. Table V shows the comparison of detection performance of the proposed model with two other models, i.e. Z. He [19] and X. Zhao [20] on DVMM dataset. We look at the Accuracy index of the proposed model and the other two models, the proposed model is at least 1% higher than [19] and [20].

TABLE V.
THE COMPARISON OF THE PROPOSED METHOD AND OTHER METHODS.

Metrics	Proposed	Z. He [19]	X. Zhao [20]
Accuracy (%)	*94.64%*	*93.55%*	*93.36%*

III. CONCLUSION AND FUTURE DEVELOPMENTS

In this article, we propose an image forgery detection model that has the splicing area changed in resolutions for two tampering methods namely copy-move and splicing. We use the convolutional neural network (CNN) for best image classification capabilities. A pre-trained VGG16 model is used to improve the model's performance. The idea such as instead of having to train the model from scratches, we reused and refined the model network parameters. The final test result shows that the model reach 94.64% accuracy.

Future developments: Although the Accuracy is high, the proposed model was still not implemented on a standard dataset and it has not been able locating where the tampering area lies, which give the team the opportunity to study and improve the model to be more effective and be able to segment those tampering region. Specifically, in the future, in addition to the VGG16 model, the team will test on other models to assess their performance and make comparison between those models in order to showcase the best result. At the same time, the team will develop a model to be able to localize the tampering region of the fake image.

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